SMART INDIA HACKATHON 2026

- **Problem Statement ID** – PS5
- **Problem Statement Title-** INTELLIGENT RESUME SCREENING & SEMANTIC RANKING
- **Theme-** AI Powered Recruitment & Intelligent Resume Matching
- **PS Category-** Software
- **Team ID-** T51
- **Team Name -** Code Hive

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PROPOSED SOLUTION :- COMPREHENSIVE AI RECRUITMENT PLATFORM

CORE COMPONENTS

- **Semantic Matching Engine :** Neural semantic model analyzing resumes and job descriptions for skills, experience, education, projects, and technical context
- **Explainable AI Layer :** Transparent reasons behind candidate rankings highlighting matched skills, missing skills, and supporting evidence
- **Intelligent Candidate Ranking :** Semantic match scoring and ranked candidate lists for quick identification of suitable applicants
- **End-to-End Pipeline :** Resume → Parsing → Semantic Embedding → Job Matching → Explainable Score → Candidate Ranking

KEY FEATURES

- Automated Resume Screening & Ranking in Real-Time
- Skill Gap & Requirement Analysis
- Explainable Match Score with Evidence
- Duplicate & Irrelevant Resume Detection
- Recruiter Dashboard with Candidate Comparison
- Job Description Analysis & Requirement Extraction
- Search & Filter Candidates using Semantic Queries
- Candidate Shortlisting & Recruitment Insights

PLATFORM ARCHITECTURE

- **Centralized Hiring Hub :** Unified application transforming unstructured resumes and job descriptions into intelligent candidate insights
- **Smart Matching :** Beyond keyword matching - understands context, technical depth, and role requirements
- **Transparency First :** All recommendations backed by clear, audit-able reasoning
- **Scalable Solution :** Handles large volumes of resumes and positions efficiently

TECHNICAL APPROACH

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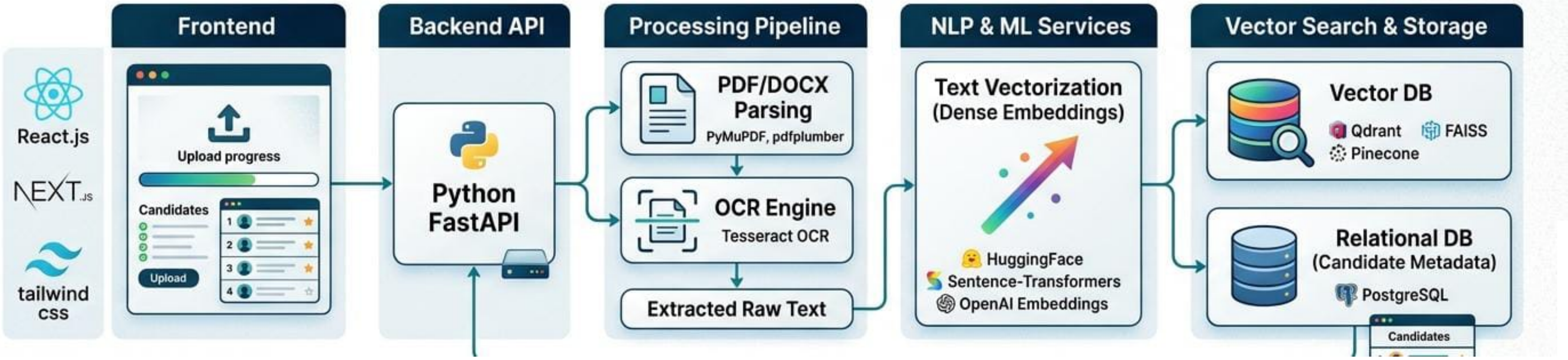
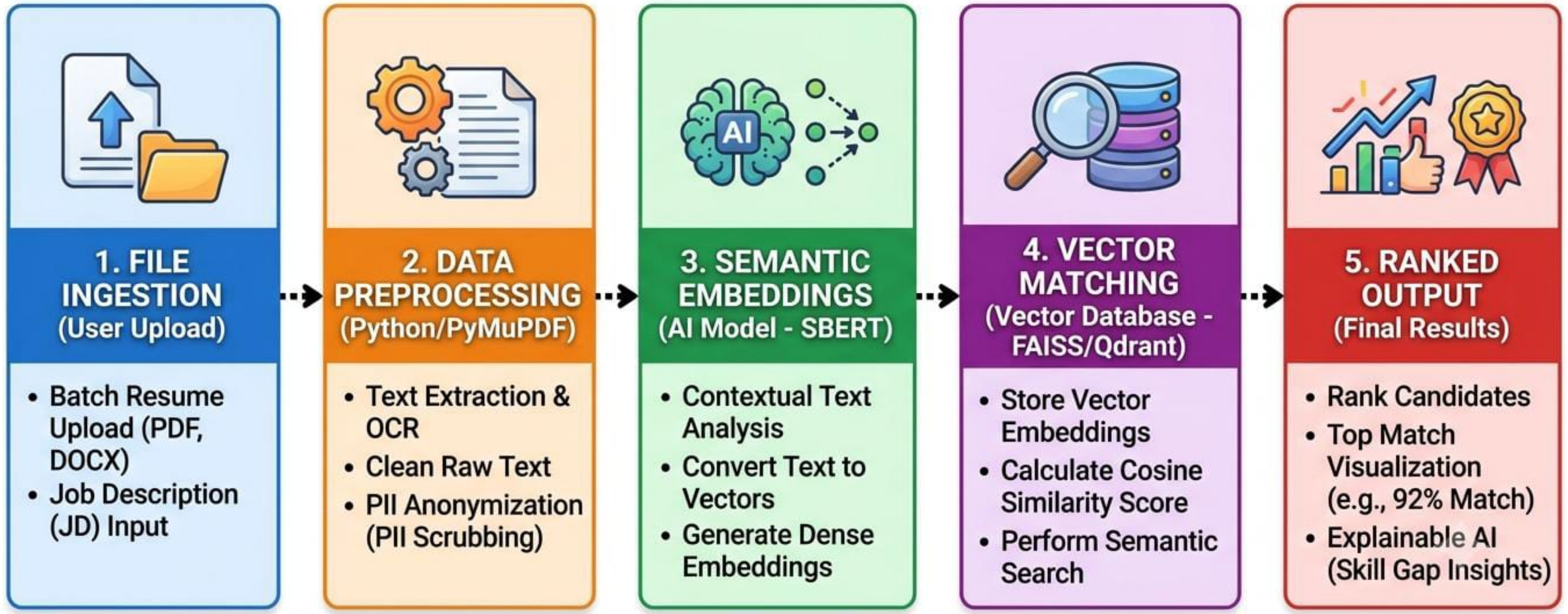
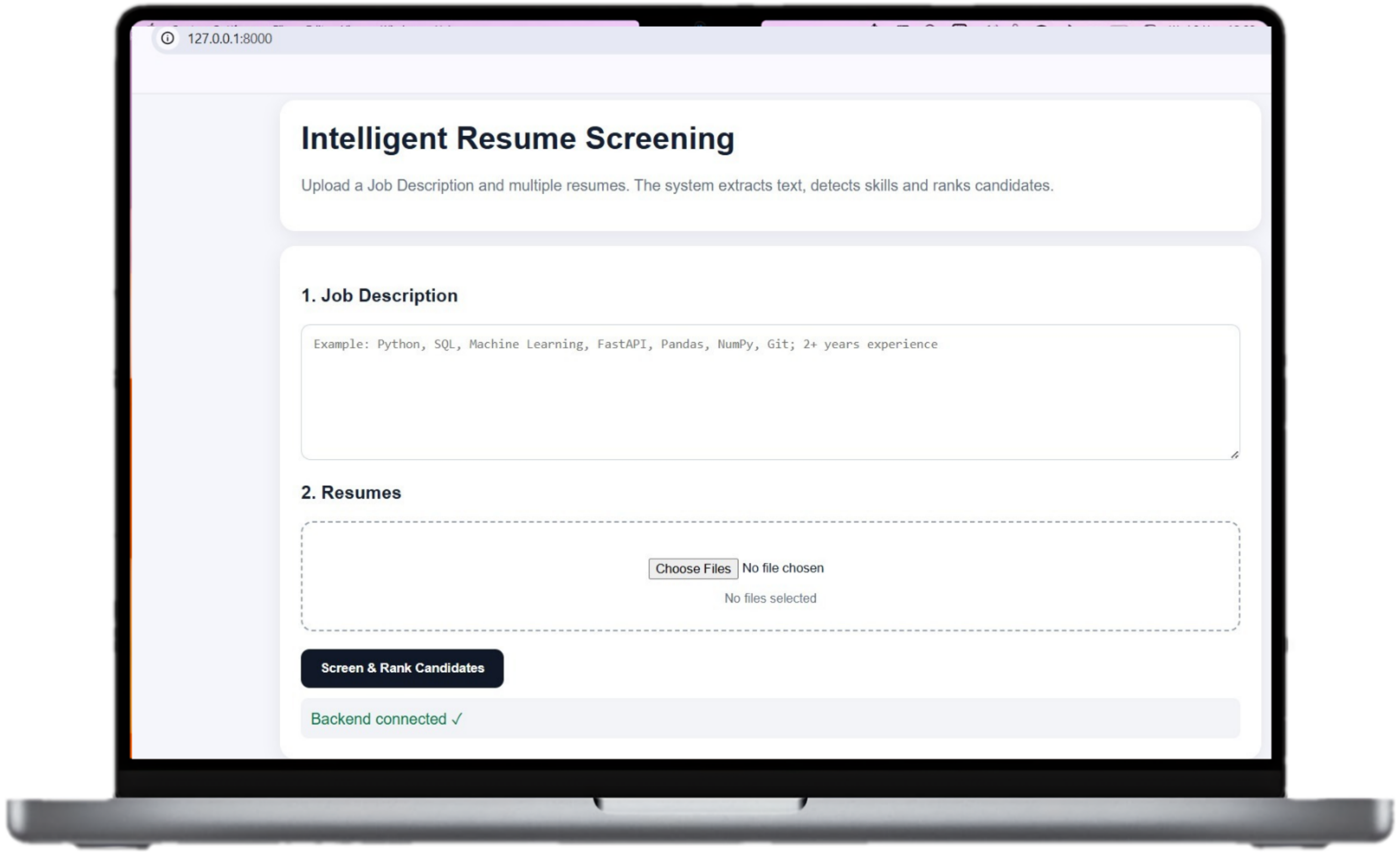
SYSTEM & TECH STACK ARCHITECTURE



SIMPLE PROCESS PIPELINE DIAGRAM





FEASIBILITY AND VIABILITY

1. FEASIBILITY ANALYSIS

Technical Feasibility :

- Advanced NLP and transformer models (SBERT, BERT)
- Converts unstructured PDF/DOCX resumes to vector embeddings
- Integrated Explainable AI (XAI) frameworks (SHAP/LIME)

Market Feasibility :

- High demand: recruiters spend only 6-7 seconds per resume scan
- Enterprises need automated, bias-free screening solutions
- Eliminates manual pipeline bottlenecks

Economic Feasibility :

- High-margin cloud-hosted SaaS model
- Reduces cost-per-hire by 60% (~$5,400+ per role)
- Automates up to 60% of manual screening effort

2. VIABILITY & SCALABILITY

Target Users :

- Corporate HR teams and technical talent acquisition specialists
- Executive search firms and recruitment agencies

Scalability Potential :

- Cloud-native architecture (AWS/Azure deployment)
- Processes tens of thousands of resumes simultaneously
- Multi-language and multi-domain expansion capability

Fairness & Compliance :

- Anonymous screening: masks demographic indicators
- Compliant with EU AI Act and regulatory standards

3. KEY CHALLENGES & SOLUTIONS

Challenge 1:
Unstructured Resume Formatting
→ **Solution:** Robust OCR and layout-aware visual parsers

Challenge 2:
Algorithmic Transparency
→ **Solution:** Integrated Explainability layer with human-readable match breakdowns

Challenge 3:
Keyword Stuffing & Fraud
→ **Solution:** Semantic vector matching evaluates context, not just keywords

4. MARKET STATISTICS

Global AI resume screening market: $2.8B in 2025

Projected market size: $12.4B by 2033

CAGR: 18.7% expansion rate

Operational Efficiency Gains:

- Automates 95% of initial candidate screening
- Cuts time-to-hire by up to 75% (36-42 days → faster)
- Boosts candidate matching accuracy by 40%

IMPACT AND BENEFITS

For Employers & Recruiters
High Efficiency:
Reduces candidate screening time by 80-90% and speeds up the hiring lifecycle.
Quality Matches:
Evaluates context and transferable skills (beyond keyword-matching) to lower early turnover.

For Job Seekers
Fair Evaluation:
Prevents instant rejections due to minor missing keywords.
Bias Reduction:
Supports blind screening (scrubbing names, demographics) to promote diversity.

Business Operations
Scalability & Savings:
Easily handles application spikes and lowers cost-per-hire.
Transparency:
Explainable AI (XAI) ensures hiring practices are compliant and clearly understood.

CANDIDATES

RESUMES

AI POWERED

RANKING

RESEARCH AND REFERENCES

KEY RESEARCH PAPERS & REFERENCES

* **conSultantBERT:**
  Link: https://doi.org/10.48550/arxiv.2109.06501

* **Smart-Hiring:**
  * Link: https://doi.org/10.48550/arXiv.1607.07657

* **CareerBERT:**
* Link: https://doi.org/10.1007/978-3-031-00123-9_3

Resume Embeddings
  * Link: https://doi.org/10.48550/arxiv.2509.09522

Demo Video Link : https://drive.google.com/file/d/18jP-B4ONZLwVNaPkmm7JX8TElULtebG6/view?usp=share_link

FUTURE SCOPE

Generative AI:
Autonomously generates customized interview questions tailored to specific skill gaps.
Agentic Workflows:
AI agents handle candidate outreach, clarify resume details, and schedule interviews.
Multimodal Evaluation:
Integrates GitHub, design portfolios, and video intros for holistic assessments.
Predictive Analytics:
Predicts long-term job fit, retention rates, and uncovers internal mobility opportunities.
Fair Hiring Frameworks:
Real-time AI fraud detection and demographic bias auditing.